

UFUG2106 Discrete Mathematics

Assignment 1

Q1 Solution

(a) Determine the Cartesian product of A and B (4 points) The Cartesian product $A \times B$ is the set of all ordered pairs (x, y) where $x \in A$ and $y \in B$. Since $|A| = 3$ and $|B| = 4$, there are $3 \times 4 = 12$ elements.

$$A \times B = \{(1, a), (1, b), (1, c), (1, d), (2, a), (2, b), (2, c), (2, d), (3, a), (3, b), (3, c), (3, d)\}.$$

(b) Define a subset $C \subseteq A \times B$ with exactly three elements (2 points)

We can choose any three distinct pairs from the set calculated in part (a). For example:

$$C = \{(1, a), (2, b), (3, c)\}.$$

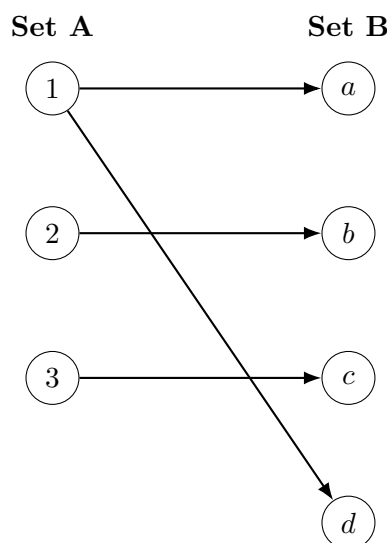
(c) Analyze the Relation R (4 points)

Given the relation

$$R = \{(1, a), (2, b), (3, c), (1, d)\}.$$

i. Directed Graph Representation (2 points)

Below is the representation of the relation R as a directed graph from Set A to Set B .



ii. Determine if R is a function (2 points)

Answer: No, R is not a function.

Justification: For a relation $f : A \rightarrow B$ to be a function, every element $x \in A$ must map to exactly one element $y \in B$. In the relation R , the element $1 \in A$ maps to two different elements in B (a and d). Since a function requires each element of A to be associated with **exactly one** element of B , but 1 is paired with both a and d in R , R fails to be a function.

Q2 Solution

(a) $(p \rightarrow q) \rightarrow r \not\equiv p \rightarrow (\sim q \vee r)$ (5 points)

p	q	r	$p \rightarrow q$	LHS: $(p \rightarrow q) \rightarrow r$	$\sim q$	$\sim q \vee r$	RHS: $p \rightarrow (\sim q \vee r)$
T	T	T	T	T	F	T	T
T	T	F	T	F	F	F	F
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	F	T	T
F	T	F	T	F	F	F	T
F	F	T	T	T	T	T	T
F	F	F	T	F	T	T	T

Conclusion: Comparing the LHS and RHS columns, the truth values differ in Row 6 and Row 8. Since the columns are not identical, the two statements are not logically equivalent.

(b) $p \rightarrow (q \rightarrow r) \equiv q \rightarrow (p \rightarrow r)$ (5 points)

p	q	r	$q \rightarrow r$	LHS: $p \rightarrow (q \rightarrow r)$	RHS: $q \rightarrow (p \rightarrow r)$
T	T	T	T	T	T
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	T	T	T
F	T	F	F	T	T
F	F	T	T	T	T
F	F	F	T	T	T

Conclusion: Comparing the LHS and RHS columns, they have identical truth values for every combination of inputs. Therefore, the two statements are logically equivalent.

Q3 Solution

(a) $((p \wedge q) \vee r) \vee (\sim p \vee (p \wedge \sim q))$ (5 points)

Solution:

$$\begin{aligned}
&= (p \wedge q) \vee r \vee (\sim p \vee (p \wedge \sim q)) \\
&= (p \wedge q) \vee r \vee ((\sim p \vee p) \wedge (\sim p \vee \sim q)) \quad (\text{Distributive Law}) \\
&= (p \wedge q) \vee r \vee (T \wedge (\sim p \vee \sim q)) \quad (\text{Negation Law}) \\
&= (p \wedge q) \vee r \vee (\sim p \vee \sim q) \quad (\text{Identity Law}) \\
&= (p \wedge q) \vee r \vee \sim (p \wedge q) \quad (\text{De Morgan's Law}) \\
&= ((p \wedge q) \vee \sim (p \wedge q)) \vee r \quad (\text{Associative/Commutative}) \\
&= T \vee r \quad (\text{Negation Law}) \\
&= T \quad (\text{Domination/Universal Bound Law})
\end{aligned}$$

(b) $(p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$ (5 points)

Solution:

$$\begin{aligned}
\text{LHS} &= p \rightarrow (\sim q \vee r) \equiv \sim p \vee \sim q \vee r, \\
\text{RHS} &= (\sim p \vee q) \rightarrow (\sim p \vee r) \quad \text{Handwritten: } p \rightarrow q \equiv \sim p \vee q \\
&= \sim (\sim p \vee q) \vee (\sim p \vee r) \quad (\text{Implication Law}) \\
&= (p \wedge \sim q) \vee \sim p \vee r \quad (\text{De Morgan's Law}) \\
&= ((\sim p \vee p) \wedge (\sim p \vee \sim q)) \vee r \quad (\text{Distributive Law}) \\
&= (T \wedge (\sim p \vee \sim q)) \vee r \\
&= \sim p \vee \sim q \vee r.
\end{aligned}$$

$$\text{Combined} = (\sim p \vee \sim q \vee r) \rightarrow (\sim p \vee \sim q \vee r) = T \quad (\text{Since } A \rightarrow A \equiv T).$$

(c) $((p \vee (q \rightarrow r)) \rightarrow p) \vee (q \wedge r)$ (5 points)

Solution:

$$\begin{aligned}
&= ((p \vee (\sim q \vee r)) \rightarrow p) \vee (q \wedge r) \quad (\text{Implication Law}) \\
&= (\sim (p \vee \sim q \vee r) \vee p) \vee (q \wedge r) \quad (\text{Implication Law}) \\
&= ((\sim p \wedge q \wedge \sim r) \vee p) \vee (q \wedge r) \quad (\text{De Morgan's Law}) \\
&= (p \vee (\sim p \wedge (q \wedge \sim r))) \vee (q \wedge r) \quad (\text{Commutative Law}) \\
&= (p \vee (q \wedge \sim r)) \vee (q \wedge r) \quad (\text{Absorption: } A \vee (\sim A \wedge B) \equiv A \vee B) \\
&= p \vee ((q \wedge \sim r) \vee (q \wedge r)) \quad (\text{Associative Law}) \\
&= p \vee (q \wedge (\sim r \vee r)) \quad (\text{Distributive Law}) \\
&= p \vee (q \wedge T) \quad (\text{Negation Law}) \\
&= p \vee q \quad (\text{Identity Law})
\end{aligned}$$

(d) $(p \wedge q) \rightarrow ((q \rightarrow r) \leftrightarrow \sim p)$ (5 points)

Solution:

$$\begin{aligned} &= \sim (p \wedge q) \vee ((\sim q \vee r) \leftrightarrow \sim p) \quad (\text{Implication Law}) \\ &= (\sim p \vee \sim q) \vee [((\sim q \vee r) \wedge \sim p) \vee (\sim (\sim q \vee r) \wedge p)] \quad (\text{Definition of } \leftrightarrow) \\ &= [\sim p \vee ((\sim q \vee r) \wedge \sim p)] \vee \sim q \vee ((q \wedge \sim r) \wedge p) \quad (\text{Rearranging}) \\ &= \sim p \vee \sim q \vee (p \wedge q \wedge \sim r) \quad (\text{Absorption: } A \vee (B \wedge A) \equiv A) \end{aligned}$$

$$\begin{aligned} \text{Let } S &= \sim p \vee \sim q \Rightarrow \sim S = p \wedge q \\ &= S \vee (\sim S \wedge \sim r) \\ &= S \vee \sim r \quad (\text{Redundancy: } A \vee (\sim A \wedge B) \equiv A \vee B) \\ &= (\sim p \vee \sim q) \vee \sim r \\ &= \sim p \vee \sim q \vee \sim r \end{aligned}$$

Q4 Solution

(a) (10 points)

Premises:

1. $\sim p \vee q \rightarrow r$
2. $s \vee \sim q$
3. $\sim t$
4. $p \rightarrow t$
5. $\sim p \wedge r \rightarrow \sim s$

Conclusion: $\sim q$

Step-by-Step Derivation:

1. $p \rightarrow t$ (Premise)
 $\sim t$ (Premise)
 $\therefore \sim p$ (Modus Tollens) (1 point)
2. $\sim p$ (from Step 1)
 $\therefore \sim p \vee q$ (Generalization) (3 point)
3. $\sim p \vee q \rightarrow r$ (Premise)
 $\sim p \vee q$ (from Step 2)
 $\therefore r$ (Modus Ponens) (5 point)
4. $\sim p$ (from Step 1)
 r (from Step 3)
 $\therefore \sim p \wedge r$ (Conjunction) (7 point)
5. $\sim p \wedge r \rightarrow \sim s$ (Premise)
 $\sim p \wedge r$ (from Step 4)
 $\therefore \sim s$ (Modus Ponens) (9 point)

6. $s \vee \sim q$ (Premise)
 $\sim s$ (from Step 5)
 $\therefore \sim q$ (Elimination) (10 point)

(b) (10 points)

Premises:

1. $(p \rightarrow q) \wedge (r \rightarrow s)$
2. $(p \rightarrow w) \wedge (s \rightarrow x)$
3. $\sim (w \wedge x)$
4. r

Conclusion: $\sim p$

Step-by-Step Derivation:

1. $(p \rightarrow q) \wedge (r \rightarrow s)$ (Premise)
 $\therefore r \rightarrow s$ (Specialization) (1 point)
2. $r \rightarrow s$ (from Step 1)
 r (Premise)
 $\therefore s$ (Modus Ponens) (2 point)
3. $(p \rightarrow w) \wedge (s \rightarrow x)$ (Premise)
 $\therefore s \rightarrow x$ (Specialization) (3 point)
4. $s \rightarrow x$ (from Step 3)
 s (from Step 2)
 $\therefore x$ (Modus Ponens) (4 point)
5. $\sim (w \wedge x)$ (Premise)
 $\therefore \sim w \vee \sim x$ (De Morgan) (5 point)
6. $\sim w \vee \sim x$ (from Step 5)
 x (from Step 4)
 $\therefore \sim w$ (Elimination) (6 point)
7. $(p \rightarrow w) \wedge (s \rightarrow x)$ (Premise)
 $\therefore p \rightarrow w$ (Specialization) (7 point)
8. $p \rightarrow w$ (from Step 7)
 $\sim w$ (from Step 6)
 $\therefore \sim p$ (Modus Tollens) (10 point)

Q5 Solution

1. Define Propositional Variables (2 point)

First, we assign variables to the atomic statements found in the problem:

- P : There is an undeclared variable.

- Q : There is a syntax error in the first five lines.
- R : There is a missing semicolon.
- S : A variable name is misspelled.

2. Translate Premises (2 point)

We translate the provided statements (a through d) into logical expressions:

- (a) $P \vee Q$
- (b) $Q \rightarrow (R \vee S)$
- (c) $\neg R$
- (d) $\neg S$

3. Step-by-Step Deduction (6 point)

- $\neg R$ (c)
 $\neg S$ (d)
 $\therefore \neg R \wedge \neg S$ (Conjunction) (2 point)
- $\neg R \wedge \neg S$ (from Step 1)
 $\therefore \neg(R \vee S)$ (De Morgan's Laws) (4 point)
- $Q \rightarrow (R \vee S)$ (b)
 $\neg(R \vee S)$ (from Step 2)
 $\therefore \neg Q$ (Modus Tollens) (7 point)
- $P \vee Q$ (a)
 $\neg Q$ (from Step 3)
 $\therefore P$ (Elimination) (10 point)

Conclusion: The mistake in the program is an **undeclared variable** (P).